

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:	28/07/2026	Duration:	60 minutes

Code of Conduct

I M.SANA FATHIMA (192371082) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M,SANA FATHIMA

Annexure A – Scenario-Based Requirement Analysis

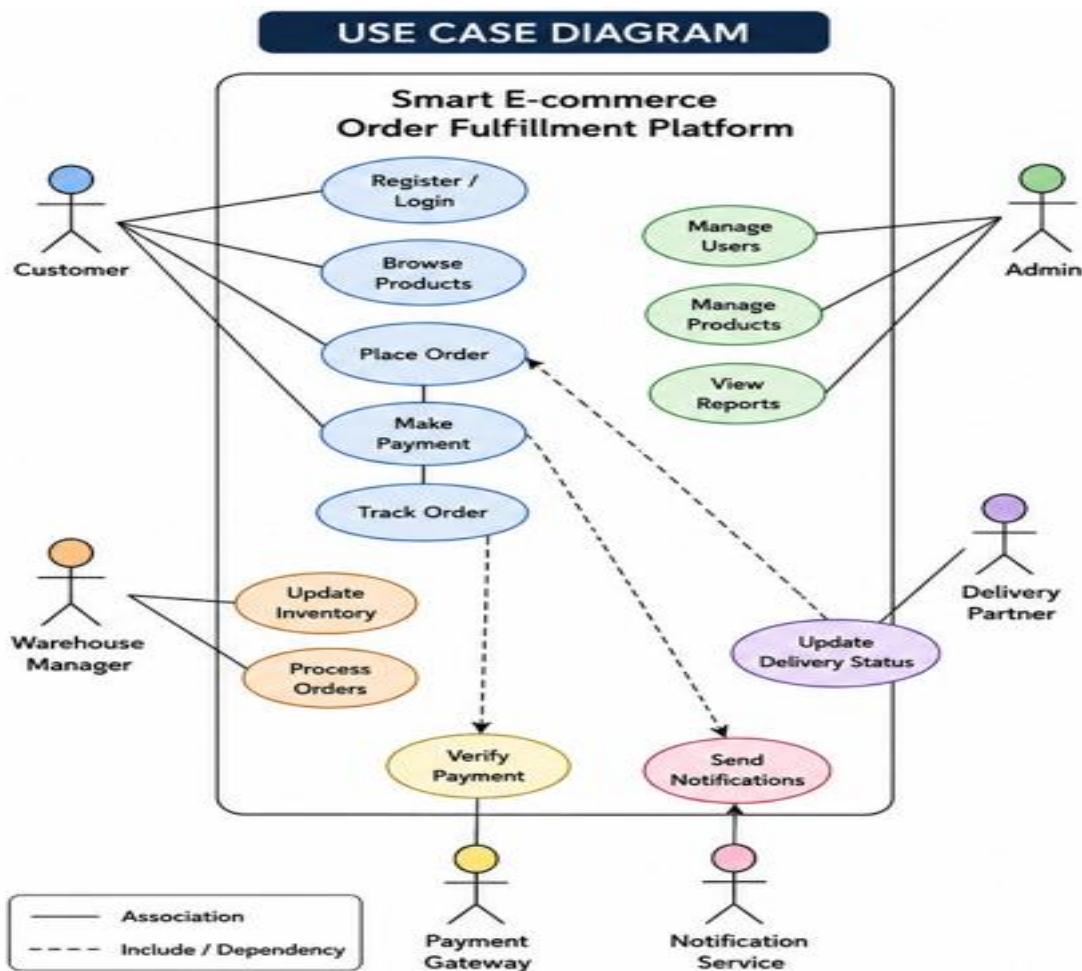
Project Title: Smart E-commerce Order Fulfillment Platform

1. Study and Analyze the Scenario

The Smart E-commerce Order Fulfillment Platform is developed for the e-commerce and logistics business domain to automate the complete order fulfillment process. In the existing system, many online businesses face challenges such as manual order processing, inaccurate inventory management, delayed shipments, poor warehouse coordination, and lack of real-time order tracking. These issues increase operational costs and reduce customer satisfaction. The proposed system aims to overcome these challenges by providing a centralized cloud-based platform that manages customer orders, inventory, warehouse allocation, payment processing, and delivery tracking. The major stakeholders include customers, administrators, warehouse managers, delivery partners, inventory managers, and payment gateway providers. The main objective of the system is to improve business efficiency, reduce processing time, optimize warehouse operations, provide accurate inventory updates, and ensure timely delivery of products while offering a better shopping experience to customers.

2. Functional and Non-Functional Requirements

- The proposed system provides several functional requirements to automate the business process. Customers can create an account, log in securely, browse products, search for items, add products to the shopping cart, place orders, make online payments, and track their order status. Administrators can manage customer accounts, products, inventory, warehouses, and reports
- The non-functional requirements ensure that the system performs efficiently and securely. The system should provide fast response time for processing customer requests and support multiple users simultaneously. It should implement secure authentication, encrypted data storage, and secure payment processing to protect customer information.



3. Actors and Use Cases

- The Smart E-commerce Order Fulfillment Platform includes both primary and secondary actors who interact with the system to perform different operations. The primary actors are the Customer, Admin, Warehouse Manager, and Delivery Partner. The secondary actors are the Payment Gateway and Notification Service. Customers interact with the system to register, log in, browse products, place orders, make payments, and track deliveries.
- Administrators manage products, users, inventory, warehouses, and business reports. Warehouse managers update inventory records and process customer orders before dispatching them. Delivery partners update delivery information and confirm product delivery. The Payment Gateway verifies secure online transactions, while the Notification Service sends order confirmations and delivery updates to customers.

4. Software Requirement Specification (SRS)

Introduction

The Smart E-commerce Order Fulfillment Platform is a cloud-based software solution developed to automate the order fulfillment process for online businesses. The system integrates order management, inventory control, warehouse allocation, payment processing, and delivery tracking into a single platform. This helps businesses improve operational efficiency and provide better customer service.

Problem Description

Many e-commerce companies still rely on manual order processing and inventory management, which often leads to delayed deliveries, stock shortages, incorrect order processing, and customer dissatisfaction. The proposed system addresses these problems by automating all major business activities and providing real-time monitoring of inventory and order status.

Scope of the System

The scope of the system includes customer registration, product management, order processing, payment verification, inventory management, warehouse allocation, delivery

tracking, customer notifications, and report generation. The system also provides administrators with business analytics to monitor sales performance and inventory status.

Functional Requirements

The system shall allow customers to register and log in securely. It shall enable customers to browse products, place orders, and make online payments. The system shall automatically update inventory after every successful order and allocate the nearest warehouse for faster delivery. It shall also track delivery status, send notifications, and generate business reports for administrators.

Non-Functional Requirements

The system shall provide high performance, data security, reliability, scalability, availability, maintainability, and a user-friendly interface. It should support multiple users simultaneously while ensuring secure storage of customer and payment information.

- Pre-condition: The customer must have registered account and be logged into the system.
- Description: The customer browses available products, selects the required items, adds them to the shopping cart, completes the online payment, and confirms the order. The system verifies the payment, updates inventory, assigns the nearest warehouse, and sends an order confirmation to the customer.
- Post-condition: The order is successfully stored in the database, inventory is updated, and the customer receives a confirmation notification.

Data Requirements

The system stores customer details, product information, order records, payment details, warehouse information, inventory records, shipment details, and notification history in a secure database to support business operations.

External Interface Requirements

The application provides a web-based user interface that can be accessed through computers and mobile devices. It connects to a MySQL database, integrates with online payment gateways, and communicates with email or SMS services to send notifications.

System Features

The major features of the system include customer registration, product browsing, secure payment processing, inventory management, warehouse allocation, delivery tracking, order notifications, report generation, and business analytics.

5. Assumptions and Constraints

The proposed system assumes that all users have internet connectivity and valid login credentials. It also assumes that inventory information is updated regularly and that payment gateway services are always available. Delivery partners are expected to update shipment information promptly. The project depends on cloud infrastructure, internet availability, project budget, and completion within the academic schedule.

6. Requirement Validation

The proposed requirements successfully address the needs of all stakeholders, including customers, administrators, warehouse managers, delivery partners, and payment service providers. The functional and non-functional requirements are complete, clearly defined, and consistent with the project objectives. The requirements do not contain ambiguity, redundancy, or conflicts. Future enhancements such as AI-based demand forecasting, chatbot support, product return management, real-time GPS tracking, and predictive inventory management can further improve the performance and functionality of the Smart E-commerce Order Fulfillment Platform.